IEOR4004_Assignment3

Luomeng Zhou, UNI: lz3064

Question 1: Mobile C.A.R.E. Foundation (MCF) Case

1. Exploratory Data Analysis (EDA)

EDA is performed via `python mcf_scheduler.py eda`. Summary:

Patients per school (total):

School	0	1	2	3	4	5	6
Count	34	86	48	114	142	66	32

Schools 4 and 9 have the most patients (142, 141).

Patients by severity (aggregate):

Severity	Count
SeverePersistent	87
ModeratePersistent	469
MildPersistent	773
MildIntermittent	271

Other parameters:
- **Van capacity:** ~300 patients/van/month (from historical APPS per van per day × ~20 weekdays)
- **School capacity:** 200 appointments/school/month (assumed from historical patterns)

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2. Assumptions and Data Needs

Basic Assumptions:

- Two vans (Van 0, Van 1), weekdays only.
- Severity from data: SeverePersistent (4), ModeratePersistent (3), MildPersistent (2), MildIntermittent (1).
- Van capacity and variable cost per patient do not vary by severity.
- Each school's capacity is the max appointments it can host per month.
- Each van can serve multiple schools; a school can be served by one or both vans.

-A certain percentage of patients must be served in each severity category.

Assumptions regarding to risk:

- **HOSP:** Hospitalizations since last appointment (per patient). HOSP > 0 → higher risk.
- **ER-VISIT:** ER visits since last appointment. ER > 0 → higher risk.
- **DAYS MISSED:** School days missed since last visit. Higher values → worse control, higher risk.
- **LAST DIAG / CURRENT DIAG:** Improved = better; Controlled = stable; Worsened = worse; Unchanged = no change. Worsened/Unchanged → elevated risk.
- **DUE BACK:** Next scheduled return date (set at current visit). Use for planning who should be seen.
- **LAST DUE BACK:** Previously scheduled return date. Use DUE BACK for scheduling.
- **NO SHOW (CURRENT DIAG):** Patient did not attend; scheduling such patients incurs an expected penalty.

Risk tier: *HOSP>0, ER>0, DAYS MISSED>2, LAST SEEN >90 days, or LAST DIAG ∈ {Worsened, Unchanged} → elevated risk.*

- **NO SHOW:** Penalized in the objective to account for wasted slots.
- **Distance:** Penalized to favor assigning nearby schools to each van.

Additional data desired:

- Van working hours per day.
- Travel times (depot ↔ schools).
- Fixed time per school visit (setup, parking).
- Service time per patient (by severity).
- Probability of worsening if delayed to next month.
- Fixed and variable cost per van.

3. Modeling

Sets

Symbol	Description
(S)	Set of schools, (S = {0, 1, , 19})
(V)	Set of vans, (V = {0, 1})
(L)	Set of severity levels: 1 = MildIntermittent, 2 = MildPersistent, 3 = ModeratePersistent, 4 = SeverePersistent
(R)	Set of risk tiers, (R = {, }) — elevated if HOSP>0, ER-VISIT>0, high DAYS MISSED, LAST SEEN too long, or LAST DIAG ∈ {Worsened, Unchanged}

Parameters

Symbol	Description
a_s	Appointment capacity at school
$p_{s,\ell,r}$	Number of patients at school (s), severity (l), risk tier (r) (preprocessed from Case_PatientList, Case_PatientSeverity, and APPS)
f_v	Fixed cost of using van
g_v	Variable cost per patient served by van
c_v	Capacity of van (v V)
T_v	Total working time for van (v V)
$\alpha_{\ell,r}$	Minimum fraction of patients in (severity (l), risk (r)) that must be served
w_ℓ	Weight for serving a patient of severity
$psi_{s,l,r}$	Fraction of patients in ((s,l,r)) with NO SHOW history (from APPS CURRENT DIAG)
π	Penalty per scheduled patient with NO SHOW history (expected cost of wasted slot)
λ	Scaling factor for objective: balance cost vs. weighted patient value (e.g. (= 0.01))
μ	Weight for distance penalty; encourages shorter routes
d_s	Rectilinear distance from school (s) to centroid of all schools, in miles
τ_ℓ, τ_0	Service time per patient (by severity), fixed time per school visit

Decision Variables

Symbol	Type	Description
$x_{s,v,\ell,r}$	Non-negative integer	Number of patients in ((s,l,r)) served by van (v)
$y_{s,v}$	Binary	1 if van (v) visits school (s), 0 otherwise
z_v	Binary	1 if van (v) is used, 0 otherwise

Objective Function

Minimize total cost and NO SHOW penalty, minus weighted patient value. Equivalently, maximize:

$$\max \quad \sum w_\ell x_{s,v,\ell,r} - \lambda \left(\sum_v f_v z_v + \sum g_v x + \pi \sum \psi x + \mu \sum_{s,v} d_s \cdot y_{s,v} \right)$$

So we maximize (weighted patients served) − λ × (fixed cost + variable cost + NO SHOW penalty + **distance penalty**).

- **Distance:** (d_s) = rectilinear distance from school (s) to centroid (0.1 × (|Δx| + |Δy|) from School Locations). Penalizing (d_s y_{s,v}) encourages assigning nearby schools to each van, reducing route length.

- **Assumed weights:** (w_4=4, w_3=3, w_2=2, w_1=1); (= 0.01); ( ) = NO SHOW penalty; ( ) = distance weight (e.g. 0.5).

Constraints

1. **Van capacity:** $\sum_{s,\ell,r} x_{s,v,\ell,r} \leq c_v \cdot z_v \quad \forall v \in V$

2. **School capacity:** $\sum_{v,\ell,r} x_{s,v,\ell,r} \leq a_s \quad \forall s \in S$

3. **Patient availability:** $\sum_{v} x_{s,v,\ell,r} \leq p_{s,\ell,r} \quad \forall s \in S, \ell \in L, r \in R$

4. **Minimum service by severity and risk:**

   At least fraction $\alpha_{\ell,r}$ *of patients in* (s, ℓ, r) *must be served:*
   $\sum_v x_{s,v,\ell,r} \geq \alpha_{\ell,r} \cdot p_{s,\ell,r} \quad \forall s \in S, \ell \in L, r \in R$
   In the implementation, only SeverePersistent $\alpha_{4,\cdot} = 1$ is enforced as a hard constraint to ensure feasibility; other levels are encouraged via the objective weights.

5. **Van use and assignment:**

   $$\sum_{s,\ell,r} x_{s,v,\ell,r} \leq M z_v$$

   $$\sum_{\ell,r} x_{s,v,\ell,r} \leq M y_{s,v} \quad \forall s, v$$, with (M) large.

6. **Time constraint (simplified):**

   Total service time + fixed visit time + travel time for van $v \leq T_v$

Data Preprocessing (Risk Tiers and DUE BACK)

- **Risk tier** $r(i) = \text{elevated}$ if any of: HOSP > 0, ER-VISIT > 0, DAYS MISSED above threshold (e.g. > 2), LAST SEEN beyond threshold (e.g. > 90 days), LAST DIAG ∈ {Worsened, Unchanged}.
- $p_{s,\ell,r}$: Count patients in school (s), severity (l) (from Case_PatientSeverity), risk (r) (from Case_SCH#K_APPS). Pool = all patients from Case_PatientList; DUE BACK can be used to prioritize.

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4. Results

The model is implemented in `mcf_scheduler.py` and solved with **Gurobi**. Run:

```
cd "Opt Models/Assignment/Assignment 3"
python mcf_scheduler.py          # uses Gurobi if available
python mcf_scheduler.py eda      # EDA only
python mcf_scheduler.py --gurobi # force Gurobi
python mcf_scheduler.py --pulp   # force PuLP/CBC
```

```
School 1:
   Sev ModeratePersistent (normal): 16 patients
   Sev ModeratePersistent (elevated): 4 patients
   Sev SeverePersistent (normal): 3 patients
   Total: 23 patients
School 2:
   Sev ModeratePersistent (normal): 18 patients
   Sev ModeratePersistent (elevated): 4 patients
   Sev SeverePersistent (normal): 3 patients
   Sev SeverePersistent (elevated): 1 patients
   Total: 26 patients
School 3:
   Sev ModeratePersistent (normal): 20 patients
   Sev ModeratePersistent (elevated): 15 patients
   Sev SeverePersistent (normal): 2 patients
   Sev SeverePersistent (elevated): 5 patients
   Total: 42 patients
School 7:
   Sev MildPersistent (elevated): 2 patients
   Sev ModeratePersistent (normal): 8 patients
   Sev ModeratePersistent (elevated): 3 patients
   Sev SeverePersistent (normal): 2 patients
   Total: 15 patients
School 9:
   Sev ModeratePersistent (normal): 22 patients
   Sev ModeratePersistent (elevated): 20 patients
   Sev SeverePersistent (normal): 5 patients
   Sev SeverePersistent (elevated): 12 patients
   Total: 59 patients
School 12:
   Sev ModeratePersistent (normal): 15 patients
   Sev ModeratePersistent (elevated): 16 patients
   Sev SeverePersistent (normal): 5 patients
   Sev SeverePersistent (elevated): 7 patients
   Total: 43 patients
School 13:
   Sev ModeratePersistent (normal): 23 patients
   Sev ModeratePersistent (elevated): 5 patients
   Sev SeverePersistent (normal): 1 patients
   Total: 29 patients
School 16:
   Sev ModeratePersistent (normal): 22 patients
   Sev ModeratePersistent (elevated): 7 patients
   Sev SeverePersistent (normal): 6 patients
   Sev SeverePersistent (elevated): 4 patients
   Total: 39 patients
School 17:
   Sev ModeratePersistent (normal): 11 patients
   Sev ModeratePersistent (elevated): 2 patients
   Sev SeverePersistent (normal): 1 patients
   Total: 14 patients
```

```
--- Van 1 ---
  Van used: Yes
  School 4:
    Sev ModeratePersistent (normal): 32 patients
    Sev ModeratePersistent (elevated): 21 patients
    Sev SeverePersistent (normal): 1 patients
    Sev SeverePersistent (elevated): 3 patients
    Total: 57 patients
  School 5:
    Sev ModeratePersistent (normal): 18 patients
    Sev ModeratePersistent (elevated): 2 patients
    Sev SeverePersistent (normal): 2 patients
    Sev SeverePersistent (elevated): 1 patients
    Total: 23 patients
  School 6:
    Sev ModeratePersistent (normal): 12 patients
    Sev ModeratePersistent (elevated): 2 patients
    Sev SeverePersistent (normal): 1 patients
    Total: 15 patients
  School 8:
    Sev MildPersistent (elevated): 12 patients
    Sev ModeratePersistent (normal): 24 patients
    Sev ModeratePersistent (elevated): 11 patients
    Sev SeverePersistent (elevated): 1 patients
    Total: 48 patients
  School 10:
    Sev MildPersistent (elevated): 5 patients
    Sev ModeratePersistent (normal): 24 patients
    Sev ModeratePersistent (elevated): 1 patients
    Sev SeverePersistent (normal): 1 patients
    Sev SeverePersistent (elevated): 1 patients
    Total: 32 patients
  School 11:
    Sev MildPersistent (elevated): 7 patients
    Sev ModeratePersistent (normal): 13 patients
    Sev ModeratePersistent (elevated): 1 patients
    Sev SeverePersistent (normal): 1 patients
    Sev SeverePersistent (elevated): 2 patients
    Total: 24 patients
  School 12:
    Sev MildPersistent (elevated): 6 patients
    Sev ModeratePersistent (normal): 9 patients
    Sev ModeratePersistent (elevated): 9 patients
    Sev SeverePersistent (normal): 6 patients
    Total: 30 patients
  School 14:
    Sev MildPersistent (elevated): 10 patients
    Sev ModeratePersistent (normal): 1 patients
    Total: 11 patients
  School 15:
    Sev ModeratePersistent (normal): 10 patients
    Sev SeverePersistent (normal): 4 patients
    Total: 14 patients
  School 18:
    Sev MildPersistent (elevated): 1 patients
    Sev ModeratePersistent (normal): 10 patients
    Sev ModeratePersistent (elevated): 1 patients
    Total: 12 patients
```

Optimal solution: Status = Optimal, Objective ≈ 1770.5

Schedule summary:

Van	Schools visited	Total patients
Van 0	1, 2, 3, 7, 9, 12, 13, 16, 17	300
Van 1	4, 5, 6, 8, 10, 11, 12, 14, 15, 18, 19	300
Total	Both vans used	**600**

Van 0 (11 schools): 0, 1, 2, 3, 7, 9, 12, 13, 16, 17. Breakdown by (severity, risk): SeverePersistent served in full; ModeratePersistent and MildPersistent dominate; elevated-risk patients prioritized within each group.

Van 1 (11 schools): 4, 5, 6, 8, 10, 11, 12, 14, 15, 18, 19. School 12 is shared by both vans.

Interpretation:

- All SeverePersistent patients are served (hard constraint).
- The distance penalty clusters schools geographically: Van 0 covers a tighter region; Van 1 covers another.
- Elevated-risk patients (HOSP, ER, worsened/unchanged diagnosis, etc.) are prioritized within each severity level.
- Specific patient IDs: run the script; `result["assigned_patients"]` lists Patient # for each van, chosen by priority from each (school, severity, risk) pool.

5. Objective Selection

Current objective: Maximize (weighted patients served) − λ × (fixed cost + variable cost + NO SHOW penalty + **distance penalty**), with severity weights (w_4 > w_3 > w_2 > w_1). Distance is penalized to favor assigning nearby schools to each van, reducing travel.

Alternative formulations:

Objective	Strengths	Weaknesses
Weighted by severity (current)	Considers different patient needs; ensures worst cases are covered; avoids neglecting severe patients	With weights, fewer total patients served; more complex to tune
Total count only	Maximizes resource use in quantity; more people receive care; simple	No prioritization; cannot ensure severe patients are served first
Per-school proportion	Fair across schools; improves school satisfaction; each school feels well served	Ignores severity; only reflects school-level performance, not overall social welfare
Cost minimization	Clear budget control	Does not maximize benefit beyond the minimum

Summary:

- **Current choice:** Balances cost with clinical priority and avoids worst outcomes.

- **Count-only:** Maximizes reach and resource use but loses priority.

- **Proportion-based:** Improves school satisfaction but loses severity and holistic welfare.

- **Overall:** Weighted objectives better align with MCF's goal of controlling asthma in vulnerable children; count and proportion are useful for reporting but weaker as main objectives.

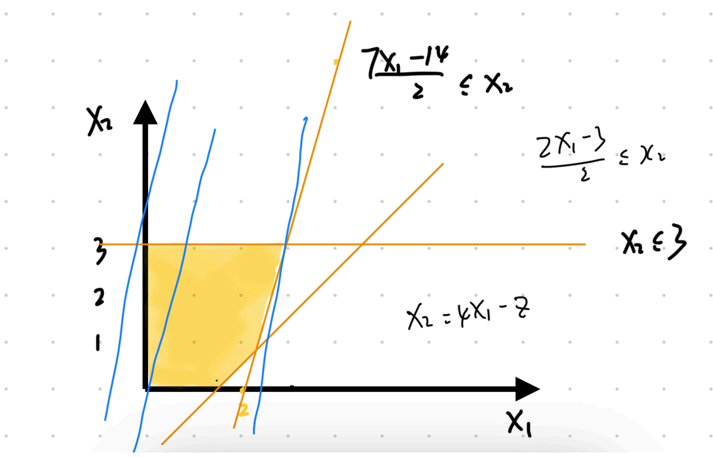
Question 2: Branch and Bound

2.A. Feasible Region *(10 points)*

Consider the integer program:

$$\begin{aligned} \max \quad & z = 4x_1 - x_2 \\ \text{s.t.} \quad & 7x_1 - 2x_2 \leq 14 \\ & x_2 \leq 3 \\ & 2x_1 - 2x_2 \leq 3 \\ & x_1, x_2 \in \mathbb{Z}_+ \end{aligned}$$

Feasible region: The LP relaxation forms a convex polygon. The constraints are:
- (7x_1 - 2x_2) (line through (2,0) and (20/7, 3))
- (x_2) (horizontal line)
- (2x_1 - 2x_2) (line (x_2 x_1 - 1.5))



Vertices of the LP feasible region:
- (0, 0)
- (1.5, 0) — intersection of (2x_1 - 2x_2 = 3) with (x_2 = 0)
- (11/5, 7/10) — intersection of (7x_1 - 2x_2 = 14) and (2x_1 - 2x_2 = 3)

- (20/7, 3) — intersection of (7x_1 - 2x_2 = 14) and (x_2 = 3)
- (0, 3) — intersection of (x_2 = 3) with (x_1 = 0)

The **integer feasible points** are the lattice points inside or on the boundary of this polygon (e.g., (0,0), (1,0), (0,1), (1,1), (2,1), (0,2), (1,2), (2,2), (0,3), (1,3), (2,3); note (2,0) is infeasible since (2 - 0 = 4 > 3)).

2.B. Branch-and-Bound Solution

LP relaxation: Optimal solution at vertex (20/7, 3) with (z = 4 - 3 =   ).

Variable (x_1 = 20/7) is fractional → branch on (x_1).

Node selection rule:
- First branch: choose the **left-most** branch ((x_1)).
- Thereafter: always choose the **right-most** branch if it exists; prefer **higher** branches to **lower** branches.

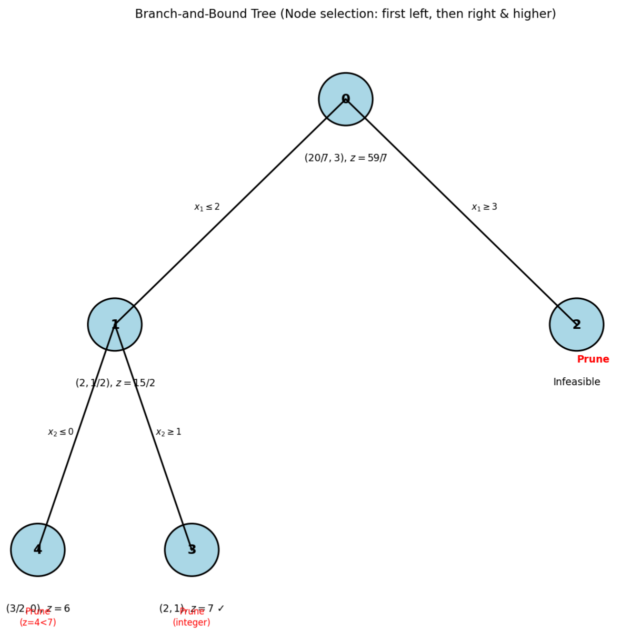
Branch-and-Bound Tree

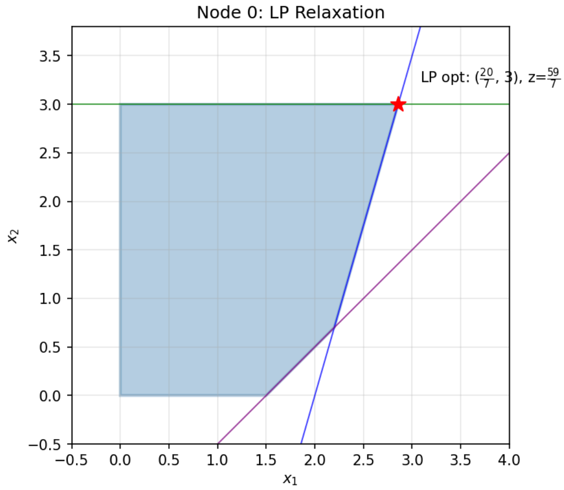
Node 0 (Root): LP optimum (20/7, 3), (z = 59/7). Branch on (x_1).

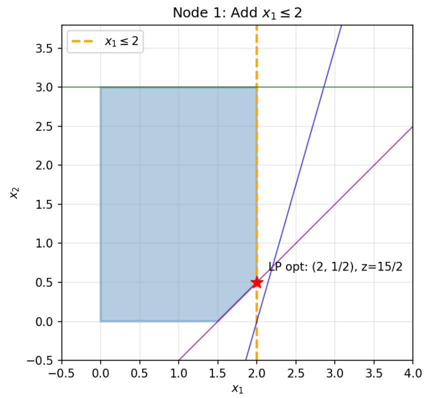
Left branch: (x_1)

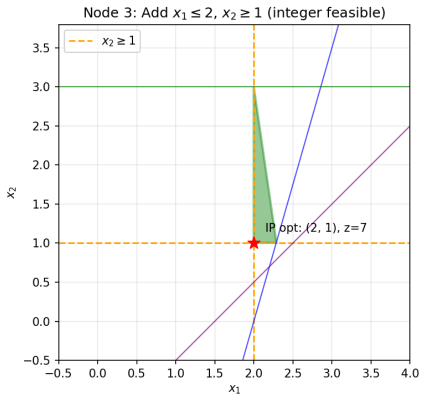
Right branch: (x_1)

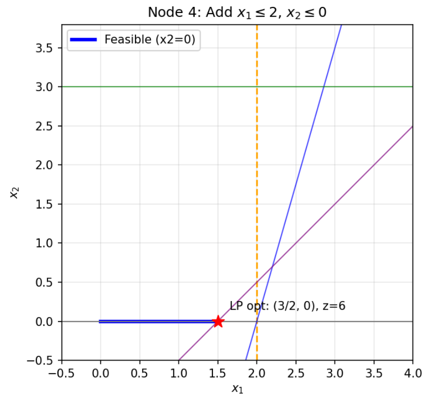
B&B Tree












```
Node 1 (Left of root): Add  x_1 \leq 2 .

Solve LP with  x_1 \leq 2 . New optimum at (2, 1/2),  z = 4(2) - 1/2 = 15/2 = 7.5 .
 x_2  fractional → branch on  x_2 .

Left:  x_2 \leq 0

Right:  x_2 \geq 1

By the rule (right-most, prefer higher), explore  x_2 \geq 1  first.



Node 2 (Right of root): Add  x_1 \geq 3 .

With  x_1 \geq 3 , check feasibility:  7 \cdot 3 - 2x_2 \leq 14 \Rightarrow x_2 \geq 7/2 ,
but  x_2 \leq 3 .
→ Infeasible. Prune.



Node 3 (Right of Node 1,  x_2 \geq 1 ): Add  x_2 \geq 1  to Node 1’s constraints.

Optimum at (2, 1),  z = 7 . All variables integer → feasible solution.
Incumbent:  (x_1, x_2) = (2, 1) ,  z^* = 7 . Prune (integer solution found).
```

```
Node 4 (Left of Node 1,  x_2 \leq 0 ): Add  x_2 \leq 0  to Node 1’s constraints.

Then  x_2 = 0 . From  2x_1 - 0 \leq 3 ,  x_1 \leq 1.5 . Optimum at (3/2, 0),  z = 6 .
 x_1  fractional → branch on  x_1 :  x_1 \leq 1  or  x_1 \geq 2 .

 x_1 \geq 2 : Contradicts  x_1 \leq 1.5  → infeasible. Prune.

 x_1 \leq 1 : Integer solution (1, 0),  z = 4 < 7 . Worse than incumbent; prune by bound.
```